

II-Probabilistic Approaches Marginalization and Conditioning

Foundations of Neuro-Symbolic AI

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Outline

- ▶ **Marginal Distributions**
- ▶ Conditional Distributions
- ▶ Bayes Theorem

Example: Being at a dentist

We are reasoning about a factored system with three variables:

- ▶ **Toothache** $i \in [2]$, whether your tooth hurts
- ▶ **Cavity** $j \in [2]$, whether there is a cavity in your tooth
- ▶ **Catch** $k \in [2]$, whether the dentist catches in your tooth

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	0.108	0.012	0.072	0.008
$\neg$ <i>cavity</i>	0.016	0.064	0.144	0.576

Formal Definition of Probability Tensors

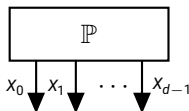
Definition (Probability Tensor)

Let there be a factored system $X_{[d]}$ defined by variables X_k taking values in $[m_k]$. A probability distribution over the states of $X_{[d]}$ is a map

$$\mathbb{P} : \bigtimes_{k \in [d]} [m_k] \rightarrow [0, \infty)$$

such that $\sum_{x_0, \dots, x_{d-1}} \mathbb{P}(x_0, \dots, x_{d-1}) = 1$.

We depict



Directed Tensors

We depict the condition, that coordinate sums are one, by directions on the legs.

Definition (Directed Tensor)

A tensor

$$\tau \in \bigotimes_{v \in \mathcal{V}} \mathbb{R}^{m_v}$$

is said to be directed with incoming variables $\mathcal{V}^{\text{in}}$ and outgoing variables $\mathcal{V}^{\text{out}}$, where $\mathcal{V} = \mathcal{V}^{\text{in}} \dot{\cup} \mathcal{V}^{\text{out}}$, when

$$\langle \{\tau\} \rangle_{[\mathcal{V}^{\text{in}}]} = \mathbb{I}^{\mathcal{V}^{\text{in}}}$$

where $\mathbb{I}^{\mathcal{V}^{\text{in}}}$ denoted the trivial tensor in $\bigotimes_{v \in \mathcal{V}^{\text{in}}} \mathbb{R}^{m_v}$ which coordinates are all 1.

Example: Marginal Distributions

What is the probability that there is a cavity?

$$\mathbb{P}^{\text{Dentist, Cavity}}[\text{Cavity}] = \sum_{j \in [2]} \sum_{k \in [2]} \mathbb{P}^{\text{Dentist}}(:, j, k) \quad (1)$$

This is called a **marginal** distribution.

Exercise

Calculate the marginal probability of **Cavity** given the probability tensor $\mathbb{P}^{\text{Dentist}}$.

	<i>toothache</i>		<i>¬toothache</i>	
	<i>catch</i>	<i>¬catch</i>	<i>catch</i>	<i>¬catch</i>
<i>cavity</i>	0.108	0.012	0.072	0.008
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Formal Definition of Marginal Distributions

Definition (Marginal Probability)

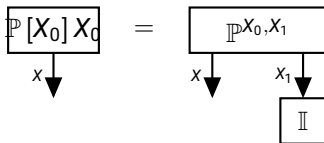
Given a distribution $\mathbb{P}$ of the categorical variables X_0 and X_1 the marginal distribution of the categorical variable X_0 is defined for each x_{X_0} as

$$\mathbb{P}[X_0 = x_{X_0}] = \sum_{x_{X_1} \in [m_{X_1}]} \mathbb{P}^{X_0=x_{X_0}, X_1=x_{X_1}}.$$

Marginal probabilities are contractions

$$\mathbb{P}^{X_0} = \langle \mathbb{P} \rangle_{[X_0]}$$

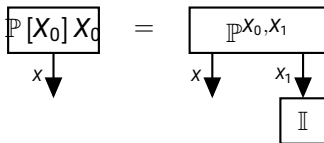
depicted by



Marginal Distributions in Contraction Formalism

Contractions are useful to

- Specify the Probability Tensor, or a decomposition of it
- Specify the variables to marginalize over as the ones left open



Directed notation preserved: Marginal probabilities are again probability distributions, since

$$\sum_{i \in [2]} \mathbb{P}^{\text{Dentist, Cavity}}[i] = \sum_{i \in [2]} \sum_{j \in [2]} \sum_{k \in [2]} \mathbb{P}_{i,j,k}^{\text{Dentist}} = 1. \quad (2)$$

Outline

- ▶ Marginal Distributions
- ▶ **Conditional Distributions**
- ▶ Bayes Theorem

Example: Conditional Distributions

What is the probability of having a cavity when having a toothache?

$$\mathbb{P}^{\text{Dentist}}[\text{Cavity}|\text{Toothache}] = \frac{\sum_{j \in [2]} \mathbb{P}^{\text{Dentist}}_{:,j,:}}{\sum_{j,k \in [2]} \mathbb{P}^{\text{Dentist}}_{:,j,k}} \quad (3)$$

This is called a **conditional** distribution.

Exercise

Calculate the probability of **Cavity** conditioned on **Toothache**.

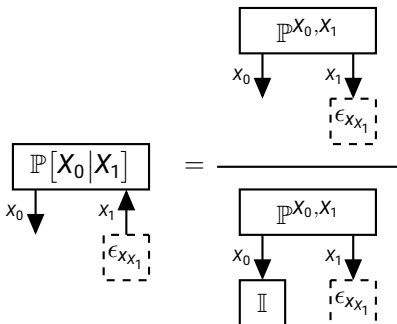
	<i>toothache</i>		<i>¬toothache</i>	
	<i>catch</i>	<i>¬catch</i>	<i>catch</i>	<i>¬catch</i>
<i>cavity</i>	0.108	0.012	0.072	0.008
<i>¬cavity</i>	0.016	0.064	0.144	0.576

Formal Definition of Conditional Distributions

Definition (Conditional Probability)

Given a distribution $\mathbb{P}$ of the categorical variables X and Y , the conditioned distribution of X is defined by

$$\mathbb{P}[X = x_X | Y = x_Y] = \frac{\mathbb{P}^{X=x_X, Y=x_Y}}{\mathbb{P}^{Y=x_Y}}.$$



Normations

Definition (Normation of Tensor Networks)

A tensor network $\tau^{\mathcal{G}}$ on variables $\mathcal{V}$ can be normed on $\mathcal{U}$, if the coordinates of no slice with respect to $\mathcal{U}$ sum to 0. Then we define the normed tensor

$$\langle \tau^{\mathcal{G}} \rangle_{[\mathcal{V}^{\text{out}} | \mathcal{V}^{\text{in}}]} \in \left(\bigotimes_{v \in \mathcal{V}^{\text{in}}} \mathbb{R}^{m_v} \right) \otimes \left(\bigotimes_{v \in \mathcal{V}^{\text{out}}} \mathbb{R}^{m_v} \right)$$

by

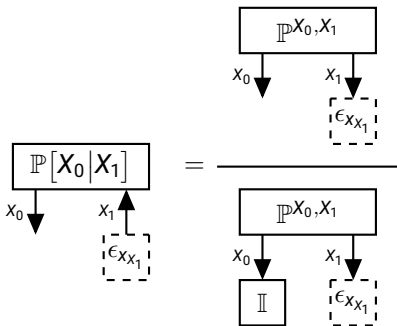
$$\langle \tau^{\mathcal{G}} \rangle_{[\mathcal{V}^{\text{out}} | \mathcal{V}^{\text{in}}]} = \sum_{x_{\mathcal{V}^{\text{in}}} \in \prod_{v \in \mathcal{V}^{\text{in}}} \mathbb{R}^{m_v}} \epsilon_{x_{\mathcal{V}^{\text{in}}}} \otimes \frac{\langle \tau^{\mathcal{G}} \cup \{\epsilon_{x_{\mathcal{V}^{\text{in}}}\} \rangle_{[\mathcal{V}^{\text{out}}]}}{\langle \tau^{\mathcal{G}} \cup \{\epsilon_{x_{\mathcal{V}^{\text{in}}}\} \rangle_{[\emptyset]}}.$$

Conditional Distributions by Normations

Conditioning is the normation

$$\mathbb{P}[X_0|X_1] = \langle \mathbb{P} \rangle_{[X_0|X_1]}$$

depicted by



The directed notation highlights **Conditions** by incoming legs and **Distributions** by outgoing legs.

Outline

- ▶ Marginal Distributions
- ▶ Conditional Distributions
- ▶ Bayes Theorem

The Bayes Theorem

The Bayes Theorem relates conditional probabilities:

Theorem (Bayes Theorem)

For any joint distribution of two categorical variables X_0 and X_1 it holds that

$$\mathbb{P}[X_0|X_1] = \frac{\mathbb{P}^{X_0, X_1}}{\mathbb{P}^{X_1}} = \frac{\mathbb{P}[X_1|X_0] \mathbb{P}^{X_0}}{\mathbb{P}^{X_1}} .$$

Bayes Theorem in the Dentist Example

Directions of Reasoning

- ▶ **Causal direction:** Toothache is caused by cavity
- ▶ **Diagnostic direction:** Cavity is probable because of toothache

Bayes Theorem allows us to reason in diagnostic direction, given an underlying causal influence:

$$\begin{aligned}\mathbb{P}^{\text{Dentist}}[\text{Cavity}|\text{Toothache}] \\ = \mathbb{P}^{\text{Dentist}}[\text{Toothache}|\text{Cavity}] \frac{\mathbb{P}^{\text{Dentist}}[\text{Cavity}]}{\mathbb{P}^{\text{Dentist}}[\text{Toothache}]}\end{aligned}$$

Contraction Calculus for Probability Tensors

Probabilistic queries can be answered by contractions

- Marginal probabilities

$$\mathbb{P}^{X_0} = \langle \mathbb{P} \rangle_{[X_0]}$$

- Conditional probabilities

$$\mathbb{P}[X_0|X_1] = \langle \mathbb{P} \rangle_{[X_0|X_1]}$$

Outlook: **Tensor network decompositions of $\mathbb{P}$** increase the execution efficiency!